

mielke-geometrodynamics: table evidence

Tables

Identifier	Page	Conf.	LaTeX source	Rendered	Image																												
mielke-geometrodynamics_TAB_001	75	—	<pre>\begin{tabular}[t]{ l l } \hline Gauge Field Terminology & Fiber Bundle Terminology \\ \hline spacetime space of phase factors symmetry or gauge group gauge transformations gauge principle classical fields gauge potential gauge field strength & base space bundle space structure group inner automorphisms of the bundle G-equivalence principle cross section of a vector bundle connection 1-form curvature 2-form \\ \hline electromagnetism electromagnetism with monopoles Dirac's monopole quantization & dynamical theory with a connection in a U(1)-bundle connection in a nontrivial U(1)-bundle over \mathbb{R}^2 \times \mathrm{S}^2 \\ \hline classification of U(1)-bundles according to the first Chern class \\ \hline Yang-Mills theory instanton number & dynamical theory with a connection in a SU(2)-bundle classification of SU(2)-bundles over S^4 according to the second Chern class \\ \hline \end{tabular}</pre>		<table><tr><th>Gauge Field Terminology</th><th>Fiber Bundle Terminology</th></tr><tr><td>spacetime</td><td>base space</td></tr><tr><td>space of phase factors</td><td>bundle space</td></tr><tr><td>symmetry or gauge group</td><td>structure group</td></tr><tr><td>gauge transformations</td><td>inner automorphisms of the bundle</td></tr><tr><td>gauge principle</td><td>G-equivalence principle</td></tr><tr><td>classical fields</td><td>cross section of a vector bundle</td></tr><tr><td>gauge potential</td><td>connection 1-form</td></tr><tr><td>gauge field strength</td><td>curvature 2-form</td></tr><tr><td>electromagnetism</td><td>dynamical theory with a connection in a U(1)-bundle</td></tr><tr><td>electromagnetism with monopoles</td><td>connection in a nontrivial U(1)-bundle over $\mathbb{R}^2 \times S^2$</td></tr><tr><td>Dirac's monopole quantization</td><td>classification of U(1)-bundles according to the first Chern class</td></tr><tr><td>Yang–Mills theory</td><td>dynamical theory with a connection in a SU(2)-bundle</td></tr><tr><td>instanton number</td><td>classification of SU(2)-bundles over S^4 according to the second Chern class</td></tr></table>	Gauge Field Terminology	Fiber Bundle Terminology	spacetime	base space	space of phase factors	bundle space	symmetry or gauge group	structure group	gauge transformations	inner automorphisms of the bundle	gauge principle	G-equivalence principle	classical fields	cross section of a vector bundle	gauge potential	connection 1-form	gauge field strength	curvature 2-form	electromagnetism	dynamical theory with a connection in a U(1)-bundle	electromagnetism with monopoles	connection in a nontrivial U(1)-bundle over $\mathbb{R}^2 \times S^2$	Dirac's monopole quantization	classification of U(1)-bundles according to the first Chern class	Yang–Mills theory	dynamical theory with a connection in a SU(2)-bundle	instanton number	classification of SU(2)-bundles over S^4 according to the second Chern class
Gauge Field Terminology	Fiber Bundle Terminology																																
spacetime	base space																																
space of phase factors	bundle space																																
symmetry or gauge group	structure group																																
gauge transformations	inner automorphisms of the bundle																																
gauge principle	G-equivalence principle																																
classical fields	cross section of a vector bundle																																
gauge potential	connection 1-form																																
gauge field strength	curvature 2-form																																
electromagnetism	dynamical theory with a connection in a U(1)-bundle																																
electromagnetism with monopoles	connection in a nontrivial U(1)-bundle over $\mathbb{R}^2 \times S^2$																																
Dirac's monopole quantization	classification of U(1)-bundles according to the first Chern class																																
Yang–Mills theory	dynamical theory with a connection in a SU(2)-bundle																																
instanton number	classification of SU(2)-bundles over S^4 according to the second Chern class																																
mielke-geometrodynamics_TAB_002	102	—	<pre>\begin{tabular}[t]{ l l l l } \hline potential & field strength & Bianchi identity & Noether identity \\ \hline \vartheta^\alpha & T^\alpha = D\vartheta^\alpha & DT^\alpha \equiv R_\gamma{}^\alpha \wedge \vartheta^\gamma & \widehat{D} \Sigma_\alpha \cong (e_\alpha \lrcorner R_\beta{}^\gamma) \wedge \Delta^\beta{}_\gamma - \frac{1}{2} (e_\alpha \lrcorner Q_{\beta\gamma}) \wedge \sigma^{\beta\gamma} \\ \Gamma_\alpha{}^\beta & R_\alpha{}^\beta = d\Gamma_\alpha{}^\beta - \Gamma_\alpha{}^\gamma \wedge \Gamma_\gamma{}^\beta & DR_\alpha{}^\beta \equiv 0 & D\Delta^\alpha{}_\beta + \vartheta^\alpha \wedge \Sigma_\beta \cong \sigma^\alpha{}_\beta \\ \hline \end{tabular}</pre>	(not rendered)	<table><tr><th>potential</th><th>field strength</th><th>Bianchi identity</th><th>Noether identity</th></tr><tr><td>ϑ^α</td><td>$T^\alpha = D\vartheta^\alpha$</td><td>$DT^\alpha \equiv R_\gamma{}^\alpha \wedge \vartheta^\gamma$</td><td>$\widehat{D} \Sigma_\alpha \cong (e_\alpha \lrcorner R_\beta{}^\gamma) \wedge \Delta^\beta{}_\gamma - \frac{1}{2} (e_\alpha \lrcorner Q_{\beta\gamma}) \wedge \sigma^{\beta\gamma}$</td></tr><tr><td>$\Gamma_\alpha{}^\beta$</td><td>$R_\alpha{}^\beta = d\Gamma_\alpha{}^\beta - \Gamma_\alpha{}^\gamma \wedge \Gamma_\gamma{}^\beta$</td><td>$DR_\alpha{}^\beta \equiv 0$</td><td>$D\Delta^\alpha{}_\beta + \vartheta^\alpha \wedge \Sigma_\beta \cong \sigma^\alpha{}_\beta$</td></tr></table>	potential	field strength	Bianchi identity	Noether identity	ϑ^α	$T^\alpha = D\vartheta^\alpha$	$DT^\alpha \equiv R_\gamma{}^\alpha \wedge \vartheta^\gamma$	$\widehat{D} \Sigma_\alpha \cong (e_\alpha \lrcorner R_\beta{}^\gamma) \wedge \Delta^\beta{}_\gamma - \frac{1}{2} (e_\alpha \lrcorner Q_{\beta\gamma}) \wedge \sigma^{\beta\gamma}$	$\Gamma_\alpha{}^\beta$	$R_\alpha{}^\beta = d\Gamma_\alpha{}^\beta - \Gamma_\alpha{}^\gamma \wedge \Gamma_\gamma{}^\beta$	$DR_\alpha{}^\beta \equiv 0$	$D\Delta^\alpha{}_\beta + \vartheta^\alpha \wedge \Sigma_\beta \cong \sigma^\alpha{}_\beta$																
potential	field strength	Bianchi identity	Noether identity																														
ϑ^α	$T^\alpha = D\vartheta^\alpha$	$DT^\alpha \equiv R_\gamma{}^\alpha \wedge \vartheta^\gamma$	$\widehat{D} \Sigma_\alpha \cong (e_\alpha \lrcorner R_\beta{}^\gamma) \wedge \Delta^\beta{}_\gamma - \frac{1}{2} (e_\alpha \lrcorner Q_{\beta\gamma}) \wedge \sigma^{\beta\gamma}$																														
$\Gamma_\alpha{}^\beta$	$R_\alpha{}^\beta = d\Gamma_\alpha{}^\beta - \Gamma_\alpha{}^\gamma \wedge \Gamma_\gamma{}^\beta$	$DR_\alpha{}^\beta \equiv 0$	$D\Delta^\alpha{}_\beta + \vartheta^\alpha \wedge \Sigma_\beta \cong \sigma^\alpha{}_\beta$																														

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																																																																																																
mielke-geometrodynamics_ TAB_003	210	—	$\begin{tabular}{t}{ l l l l l l }\hline & \text{Objects} & & \text{(p)-form} & \text{Components} & \text{(\mathrm{n}=4)} & 3 & 2 \\ \hline & \vartheta^\alpha & \text{Vector} & 1 & n^2 & 16 & 9 & 1 \\ & \Gamma_\alpha^\star & \text{Vector} & 1 & n^2 & 16 & 9 & 1 \\ & T^\alpha & \text{Vector} & 2 & n^2(n-1)/2 & 24 & 9 & 2 \\ & R^{\alpha\beta} & \text{Bivector} & 2 & n^2(n-1)^2/4 & 36 & 9 & 1 \\ & \Sigma_\alpha & \text{Vector} & n-1 & n^2 & 16 & 9 & 4 \\ & \tau_{\alpha\beta} & \text{Bivector} & n-1 & n^2(n-1)/2 & 24 & 9 & 2 \\ & \eta_\alpha & \text{Vector} & n-1 & n^2 & 16 & 9 & 4 \\ \hline \end{tabular}$	<table><tr><th></th><th>Objects</th><th><i>p</i>-form</th><th>Components</th><th>n = 4</th><th>3</th><th>2</th></tr><tr><td>ϑ^α</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>$\Gamma_\alpha^\star$</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>T^α</td><td>Vector</td><td>2</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>$R^{\alpha\beta}$</td><td>Bivector</td><td>2</td><td>$n^2(n-1)^2/4$</td><td>36</td><td>9</td><td>1</td></tr><tr><td>Σ_α</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr><tr><td>$\tau_{\alpha\beta}$</td><td>Bivector</td><td>$n-1$</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>η_α</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr></table>		Objects	<i>p</i> -form	Components	n = 4	3	2	ϑ^α	Vector	1	n^2	16	9	1	$\Gamma_\alpha^\star$	Vector	1	n^2	16	9	1	T^α	Vector	2	$n^2(n-1)/2$	24	9	2	$R^{\alpha\beta}$	Bivector	2	$n^2(n-1)^2/4$	36	9	1	Σ_α	Vector	$n-1$	n^2	16	9	4	$\tau_{\alpha\beta}$	Bivector	$n-1$	$n^2(n-1)/2$	24	9	2	η_α	Vector	$n-1$	n^2	16	9	4	<table><tr><th></th><th>Objects</th><th><i>p</i>-form</th><th>Components</th><th>n = 4</th><th>3</th><th>2</th></tr><tr><td>ϑ^α</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>$\Gamma_\alpha^\star$</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>T^α</td><td>Vector</td><td>2</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>$R^{\alpha\beta}$</td><td>Bivector</td><td>2</td><td>$n^2(n-1)^2/4$</td><td>36</td><td>9</td><td>1</td></tr><tr><td>Σ_α</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr><tr><td>$\tau_{\alpha\beta}$</td><td>Bivector</td><td>$n-1$</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>η_α</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr></table>		Objects	<i>p</i> -form	Components	n = 4	3	2	ϑ^α	Vector	1	n^2	16	9	1	$\Gamma_\alpha^\star$	Vector	1	n^2	16	9	1	T^α	Vector	2	$n^2(n-1)/2$	24	9	2	$R^{\alpha\beta}$	Bivector	2	$n^2(n-1)^2/4$	36	9	1	Σ_α	Vector	$n-1$	n^2	16	9	4	$\tau_{\alpha\beta}$	Bivector	$n-1$	$n^2(n-1)/2$	24	9	2	η_α	Vector	$n-1$	n^2	16	9	4
	Objects	<i>p</i> -form	Components	n = 4	3	2																																																																																																															
ϑ^α	Vector	1	n^2	16	9	1																																																																																																															
$\Gamma_\alpha^\star$	Vector	1	n^2	16	9	1																																																																																																															
T^α	Vector	2	$n^2(n-1)/2$	24	9	2																																																																																																															
$R^{\alpha\beta}$	Bivector	2	$n^2(n-1)^2/4$	36	9	1																																																																																																															
Σ_α	Vector	$n-1$	n^2	16	9	4																																																																																																															
$\tau_{\alpha\beta}$	Bivector	$n-1$	$n^2(n-1)/2$	24	9	2																																																																																																															
η_α	Vector	$n-1$	n^2	16	9	4																																																																																																															
	Objects	<i>p</i> -form	Components	n = 4	3	2																																																																																																															
ϑ^α	Vector	1	n^2	16	9	1																																																																																																															
$\Gamma_\alpha^\star$	Vector	1	n^2	16	9	1																																																																																																															
T^α	Vector	2	$n^2(n-1)/2$	24	9	2																																																																																																															
$R^{\alpha\beta}$	Bivector	2	$n^2(n-1)^2/4$	36	9	1																																																																																																															
Σ_α	Vector	$n-1$	n^2	16	9	4																																																																																																															
$\tau_{\alpha\beta}$	Bivector	$n-1$	$n^2(n-1)/2$	24	9	2																																																																																																															
η_α	Vector	$n-1$	n^2	16	9	4																																																																																																															
mielke-geometrodynamics_ TAB_004	275	—	$\begin{tabular}{t}{ l l l }\hline & \text{Spin} & \text{Gravitational} & \text{YM anomaly} \\ \hline & 1/2 & 1 & 1 \\ & 3/2 & -21 & 3 \\ \hline \end{tabular}$	<table><tr><th>Spin</th><th>Gravitational</th><th>YM anomaly</th></tr><tr><td>1/2</td><td>1</td><td>1</td></tr><tr><td>3/2</td><td>-21</td><td>3</td></tr></table>	Spin	Gravitational	YM anomaly	1/2	1	1	3/2	-21	3	<table><tr><th>Spin</th><th>Gravitational</th><th>YM anomaly</th></tr><tr><td>1/2</td><td>1</td><td>1</td></tr><tr><td>3/2</td><td>-21</td><td>3</td></tr></table>	Spin	Gravitational	YM anomaly	1/2	1	1	3/2	-21	3																																																																																														
Spin	Gravitational	YM anomaly																																																																																																																			
1/2	1	1																																																																																																																			
3/2	-21	3																																																																																																																			
Spin	Gravitational	YM anomaly																																																																																																																			
1/2	1	1																																																																																																																			
3/2	-21	3																																																																																																																			

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																																																												
mielke-geometrodynamics_364 TAB_005		—	$\begin{tabular}[t]{l l } \hline := & : \text{Equal by definition} \\ \hline \equiv & : \text{Identically equal} \\ \hline \approx & : \text{Isomorphic} \\ \hline \simeq & : \text{Isomorphic} \\ \hline \sim & : \text{Approximately equal} \\ \hline \sim & : \text{Asymptotic expansion} \\ \hline \times & : \text{Topological product} \\ \hline \otimes & : \text{Direct product} \\ \hline \oplus & : \text{Direct sum} \\ \hline \otimes & : \text{Semidirect product} \\ \hline \mathrm{G}_\circ & : \text{Connected component of the group} \\ \hline \tilde{\mathrm{G}} & : \text{Simply connected covering group} \\ \hline [\mathrm{A}, \mathrm{B}] := \mathrm{AB} - \mathrm{BA} & : \text{Commutator} \\ \hline \{\mathrm{A}, \mathrm{B}\} := \mathrm{AB} + \mathrm{BA} & : \text{Anticommutator} \\ \hline \mathrm{A}^* = \mathrm{C} \mathrm{A} & : \text{Complex-conjugate matrix} \\ \hline \mathrm{A}^+ = \mathrm{A}^{*\mathrm{T}} & : \text{Hermitian adjoint matrix} \\ \hline \pi^{-1}(\mathrm{N}) & : \text{Original domain of N with respect to the mapping } \pi : \mathrm{M} \rightarrow \mathrm{N} \\ \hline \mathrm{T}_{(\mu\nu)} := \frac{1}{2}(\mathrm{T}_{\mu\nu} + \mathrm{T}_{\nu\mu}) & : \text{Symmetrized second-rank tensor} \\ \hline \mathrm{T}_{[\mu\nu]} := \frac{1}{2}(\mathrm{T}_{\mu\nu} - \mathrm{T}_{\nu\mu}) & : \text{Antisymmetrized second-rank tensor} \\ \hline \end{tabular}$	<table><tr><td>:=</td><td>: Equal by definition</td></tr><tr><td>≡</td><td>: Identically equal</td></tr><tr><td>≈</td><td>: Isomorphic</td></tr><tr><td>≃</td><td>: Isomorphic</td></tr><tr><td>∼</td><td>: Approximately equal</td></tr><tr><td>∼</td><td>: Asymptotic expansion</td></tr><tr><td>×</td><td>: Topological product</td></tr><tr><td>⊗</td><td>: Direct product</td></tr><tr><td>⊕</td><td>: Direct sum</td></tr><tr><td>⊗</td><td>: Semidirect product</td></tr><tr><td>G_◦</td><td>: Connected component of the group</td></tr><tr><td>G̃</td><td>: Simply connected covering group</td></tr><tr><td>[A,B] := AB − BA</td><td>: Commutator</td></tr><tr><td>{A,B} := AB + BA</td><td>: Anticommutator</td></tr><tr><td>A* = C A</td><td>: Complex-conjugate matrix</td></tr><tr><td>A⁺ = A[*] ^T</td><td>: Hermitian adjoint matrix</td></tr><tr><td>π^{−1}(N)</td><td>: Original domain of N with respect to the mapping π : M → N</td></tr><tr><td>T_(μν) := $\frac{1}{2}$ (T_{μν} + T_{νμ})</td><td>: Symmetrized second-rank tensor</td></tr><tr><td>T_[μν] := $\frac{1}{2}$ (T_{μν} − T_{νμ})</td><td>: Antisymmetrized second-rank tensor</td></tr></table>	:=	: Equal by definition	≡	: Identically equal	≈	: Isomorphic	≃	: Isomorphic	∼	: Approximately equal	∼	: Asymptotic expansion	×	: Topological product	⊗	: Direct product	⊕	: Direct sum	⊗	: Semidirect product	G _◦	: Connected component of the group	G̃	: Simply connected covering group	[A,B] := AB − BA	: Commutator	{A,B} := AB + BA	: Anticommutator	A* = C A	: Complex-conjugate matrix	A ⁺ = A [*] ^T	: Hermitian adjoint matrix	π ^{−1} (N)	: Original domain of N with respect to the mapping π : M → N	T _(μν) := $\frac{1}{2}$ (T _{μν} + T _{νμ})	: Symmetrized second-rank tensor	T _[μν] := $\frac{1}{2}$ (T _{μν} − T _{νμ})	: Antisymmetrized second-rank tensor	<table><tr><td>:=</td><td>: Equal by definition</td></tr><tr><td>≡</td><td>: Identically equal</td></tr><tr><td>≈</td><td>: Isomorphic</td></tr><tr><td>≃</td><td>: Isomorphic</td></tr><tr><td>∼</td><td>: Approximately equal</td></tr><tr><td>∼</td><td>: Asymptotic expansion</td></tr><tr><td>×</td><td>: Topological product</td></tr><tr><td>⊗</td><td>: Direct product</td></tr><tr><td>⊕</td><td>: Direct sum</td></tr><tr><td>⊗</td><td>: Semidirect product</td></tr><tr><td>G_◦</td><td>: Connected component of the group</td></tr><tr><td>G̃</td><td>: Simply connected covering group</td></tr><tr><td>[A,B] := AB − BA</td><td>: Commutator</td></tr><tr><td>{A,B} := AB + BA</td><td>: Anticommutator</td></tr><tr><td>A* = C A</td><td>: Complex-conjugate matrix</td></tr><tr><td>A⁺ = A[*] ^T</td><td>: Hermitian adjoint matrix</td></tr><tr><td>π^{−1}(N)</td><td>: Original domain of N with respect to the mapping π : M → N</td></tr><tr><td>T_(μν) := $\frac{1}{2}$ (T_{μν} + T_{νμ})</td><td>: Symmetrized second-rank tensor</td></tr><tr><td>T_[μν] := $\frac{1}{2}$ (T_{μν} − T_{νμ})</td><td>: Antisymmetrized second-rank tensor</td></tr></table>	:=	: Equal by definition	≡	: Identically equal	≈	: Isomorphic	≃	: Isomorphic	∼	: Approximately equal	∼	: Asymptotic expansion	×	: Topological product	⊗	: Direct product	⊕	: Direct sum	⊗	: Semidirect product	G _◦	: Connected component of the group	G̃	: Simply connected covering group	[A,B] := AB − BA	: Commutator	{A,B} := AB + BA	: Anticommutator	A* = C A	: Complex-conjugate matrix	A ⁺ = A [*] ^T	: Hermitian adjoint matrix	π ^{−1} (N)	: Original domain of N with respect to the mapping π : M → N	T _(μν) := $\frac{1}{2}$ (T _{μν} + T _{νμ})	: Symmetrized second-rank tensor	T _[μν] := $\frac{1}{2}$ (T _{μν} − T _{νμ})	: Antisymmetrized second-rank tensor
:=	: Equal by definition																																																																																
≡	: Identically equal																																																																																
≈	: Isomorphic																																																																																
≃	: Isomorphic																																																																																
∼	: Approximately equal																																																																																
∼	: Asymptotic expansion																																																																																
×	: Topological product																																																																																
⊗	: Direct product																																																																																
⊕	: Direct sum																																																																																
⊗	: Semidirect product																																																																																
G _◦	: Connected component of the group																																																																																
G̃	: Simply connected covering group																																																																																
[A,B] := AB − BA	: Commutator																																																																																
{A,B} := AB + BA	: Anticommutator																																																																																
A* = C A	: Complex-conjugate matrix																																																																																
A ⁺ = A [*] ^T	: Hermitian adjoint matrix																																																																																
π ^{−1} (N)	: Original domain of N with respect to the mapping π : M → N																																																																																
T _(μν) := $\frac{1}{2}$ (T _{μν} + T _{νμ})	: Symmetrized second-rank tensor																																																																																
T _[μν] := $\frac{1}{2}$ (T _{μν} − T _{νμ})	: Antisymmetrized second-rank tensor																																																																																
:=	: Equal by definition																																																																																
≡	: Identically equal																																																																																
≈	: Isomorphic																																																																																
≃	: Isomorphic																																																																																
∼	: Approximately equal																																																																																
∼	: Asymptotic expansion																																																																																
×	: Topological product																																																																																
⊗	: Direct product																																																																																
⊕	: Direct sum																																																																																
⊗	: Semidirect product																																																																																
G _◦	: Connected component of the group																																																																																
G̃	: Simply connected covering group																																																																																
[A,B] := AB − BA	: Commutator																																																																																
{A,B} := AB + BA	: Anticommutator																																																																																
A* = C A	: Complex-conjugate matrix																																																																																
A ⁺ = A [*] ^T	: Hermitian adjoint matrix																																																																																
π ^{−1} (N)	: Original domain of N with respect to the mapping π : M → N																																																																																
T _(μν) := $\frac{1}{2}$ (T _{μν} + T _{νμ})	: Symmetrized second-rank tensor																																																																																
T _[μν] := $\frac{1}{2}$ (T _{μν} − T _{νμ})	: Antisymmetrized second-rank tensor																																																																																